

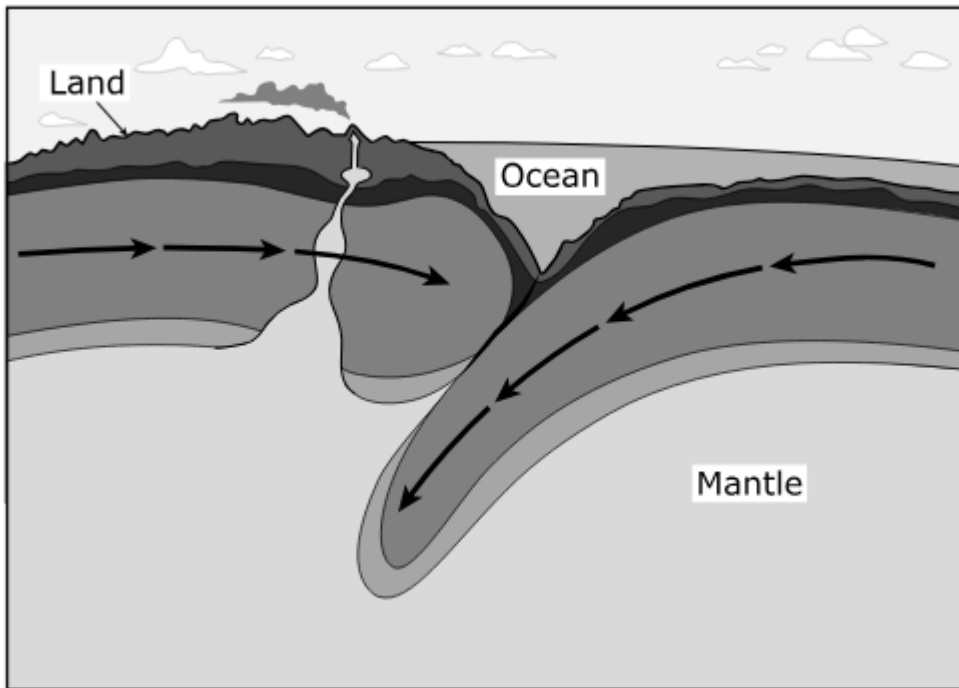
Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 8
Practice Items



The diagram shows a boundary between two tectonic plates.



According to the diagram, which feature is **most likely** being developed?

- A. a transform fault, because these plates are both moving sideways
- B. an oceanic trench, because one plate is subducting beneath the other
- C. a mountain range, because both of these continental plates are converging
- D. a mid-ocean ridge, because a mountain range is being formed by the plates separating

Before the arrival of a new bird species, beetle pairs had produced both green and brown offspring in approximately equal proportions. The main diet of a bird species in a forest area is a population of beetles. The beetles are found on tree trunks. The birds catch the green beetles more easily than the brown ones.

Initial Representation of the Beetle Population	Beetle Being Eaten by Predator	Final Representation of the Beetle Population
		

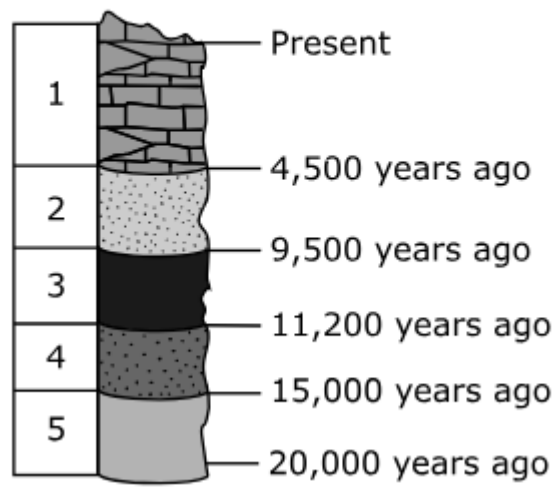
Key	
	Green Beetle
	Brown Beetle

Which consequence will **most likely** occur due to the introduction of the new bird species?

- M. The green beetles will disappear completely, and the parent beetles of both colors will now produce only brown beetles.
- P. The birds will eat only brown beetles, and the green beetles will immediately learn how to hide in the tree leaves.
- R. The number of brown beetles will increase over time, and the number of green beetles will decrease over time.
- S. The birds will learn to prefer brown beetles, and the number of green beetles on the tree trunks will increase.

Scientists removed a column-shaped core of material from the bottom of Battle Ground Lake in Washington State. Pollen from nearby plants and trees became trapped and accumulated in the sediment layers each year. These tiny pollen grains are preserved like fossils in the sediment and can be used to identify the plants growing in the area of the lake throughout its history. A model of a core sample and a data table are shown.

Model Core Sample of Battle Ground Lake, Washington



	Sediment Layer	Primary Pollen Type in Layer	Preferred Climate
shallow	1	Red cedar	Mild
	2	Mixed meadow species	Mild
	3	Alder tree	Cool
	4	Engelmann spruce	Cold
deep	5	Alpine sedges	Cold

Which conclusion is **best** supported by the information in the data table?

- A. The climate was uniformly mild over the 20,000-year period.
- B. The climate became milder over the 20,000-year period.
- C. The climate became colder over the 20,000-year period.
- D. The climate was uniformly cool over the 20,000-year period.

The chart compares the structures of four organisms.

Bird and Dinosaur Comparison

	Archaeopteryx	Sparrow	Ornithischian Dinosaur	Allosaurid
Length of Limbs	Arms longer than legs	Arms longer than legs	Legs longer than arms	Legs longer than arms
Feathers	Present	Present	Absent	Absent
Wishbone	Present	Present	Absent	Present
Number of Digits on Hand	3	3	4	3
Skeletal Structure of Hand				

Based on the chart, which organism is **least closely** related to the sparrow?

- M. Archaeopteryx
- P. Sparrow
- R. Ornithischian Dinosaur
- S. Allosaurid

Farmers have been cultivating varieties of peaches for many years. Recently, scientists have learned how to genetically modify peaches to increase their ranges of size, sweetness, and other characteristics. The table compares an average cultivated peach to an average genetically modified peach.

Cultivated Peaches vs. Genetically Modified Peaches

Peach Type	Average Diameter of Fruit (cm)	Average Diameter of Seed (cm)	Edible Part of Fruit (%)	Seed Part of Fruit (%)
Cultivated	6.5	2	70	30
Genetically Modified	10	1	90	10

Based on the table, which of these **best** describes an advantage to humans that a genetically modified peach has over a cultivated peach?

- A. Genetically modified peaches have more edible fruit than cultivated peaches.
- B. Genetically modified peaches have larger seeds than cultivated peaches.
- C. The size of genetically modified peaches makes them easier to transport to grocery stores than cultivated peaches.
- D. Genetically modified peaches cross with other fruits better than cultivated peaches and make different varieties of fruit available.

A student is alone in a school gymnasium. They stand near a door that has windows in it and then shout a few words.

Which statement provides the **best** evidence that the gymnasium reflects both light waves and sound waves?

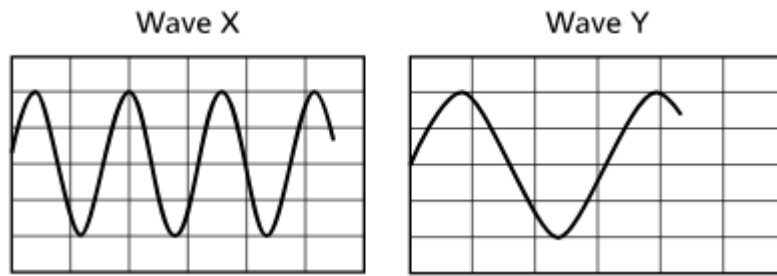
- M. The student can see marks on the gymnasium floor and hear their own voice.
- P. The student can see through the window glass and hear their own voice.
- R. The student can see their image in the window glass and hear an echo.
- S. The student can see outside through the window glass and hear an echo.

Mount St. Helens is a volcano in the state of Washington that erupted violently in 1980. Superheated gas spread quickly over eight miles around the mountain. Shock waves and ash spread over 595 square kilometers. One animal that lives on Mount St. Helens is the pocket gopher. Pocket gophers live in underground burrows.

Which statement explains the **most likely** effect that the rapid spreading of hot gas had on the pocket gopher population?

- A. A small number of the gopher population survived because the ash destroyed their habitat.
- B. The gophers were unaffected because they migrated to another mountain range.
- C. Half of the gopher population survived by hiding in the trees.
- D. Most of the gopher population was protected by hiding in their underground habitats.

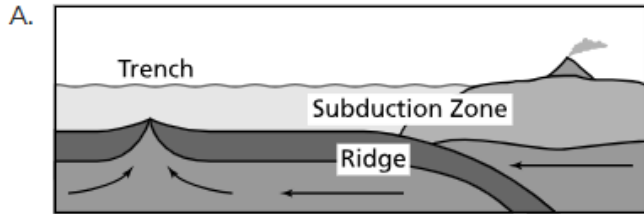
The graphs show two electromagnetic waves.



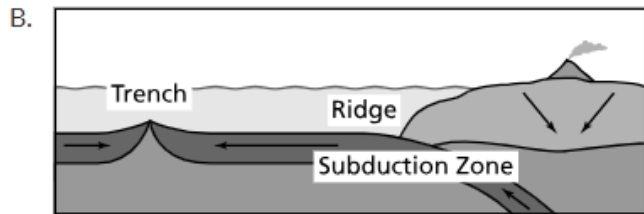
Which statement **best** compares Wave X to Wave Y?

- M. Wave X has a larger amplitude than Wave Y.
- P. Wave Y has a shorter wavelength than Wave X.
- R. Wave Y has a greater speed than Wave X.
- S. Wave X has a higher frequency than Wave Y.

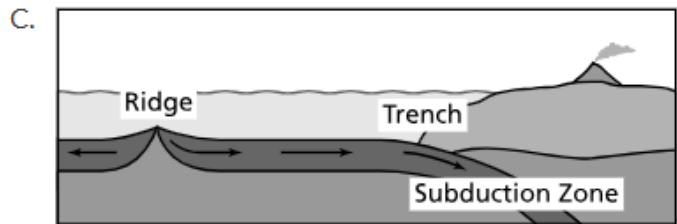
Which of these models **best** explains plate movement?



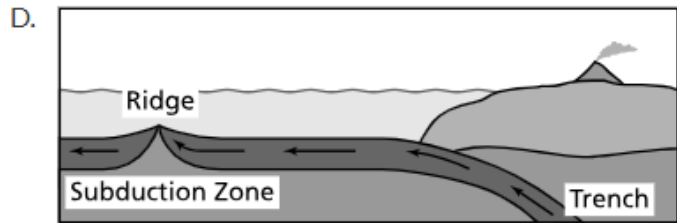
A plate pushes another plate in a subduction zone. A ridge is formed when the plate moves along a path toward a trench.



The weight from a volcano pushes one plate through a ridge at a subduction zone. The plate moves closer to another plate and a trench is formed.

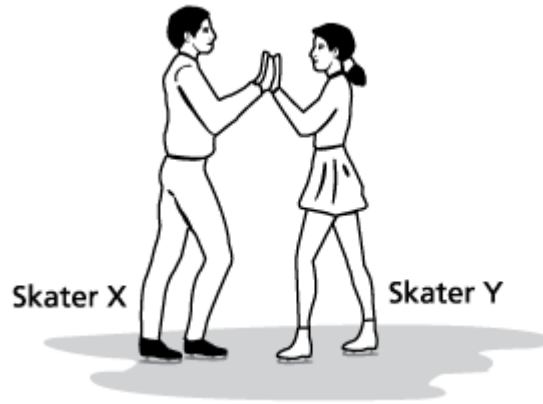


Two plates separate at a ridge. A trench is formed when one plate pushes under another plate.



One plate is pushed by a trench toward another plate at a subduction zone. The plates move closer together and a ridge is formed.

At first, the two skaters in the picture are motionless. Then they push their hands against each other with equal amounts of force.



Which of these is the **most likely** result of the skaters' action?

- M. One skater moves forward while the other skater remains in place.
- P. One skater moves backward while the other skater remains in place.
- R. Both skaters move away from each other.
- S. Both skaters move toward each other.

The questions on pages 13 - 15 refer to the passage(s) and image(s) shown.

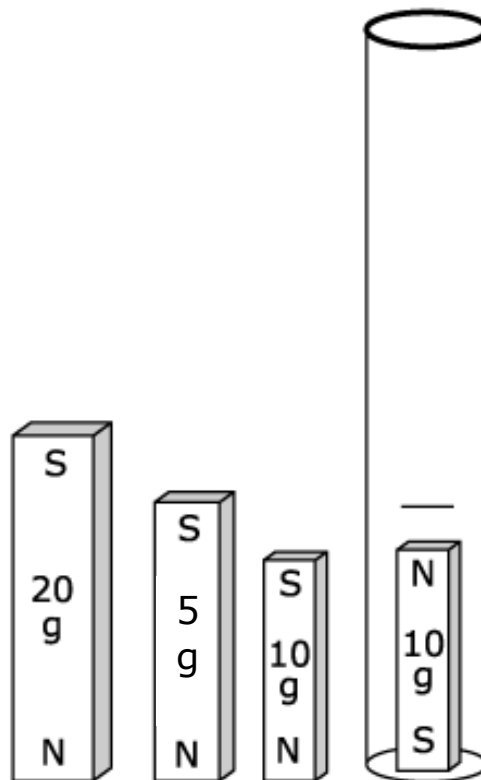
Magnets in a Tube – Part 1

Students designed an experiment to study how pairs of magnets interact when the magnets are placed near each other. The students have four bar magnets:

- 1 5-gram (5 g) magnet
- 2 10-gram (10 g) magnets
- 1 20-gram (20 g) magnet

Each magnet exerts the same amount of magnetic force. The students place one 10 g magnet inside a plastic tube so that it rests on the bottom of the tube with its north pole facing up. Figure 1 shows the students' setup.

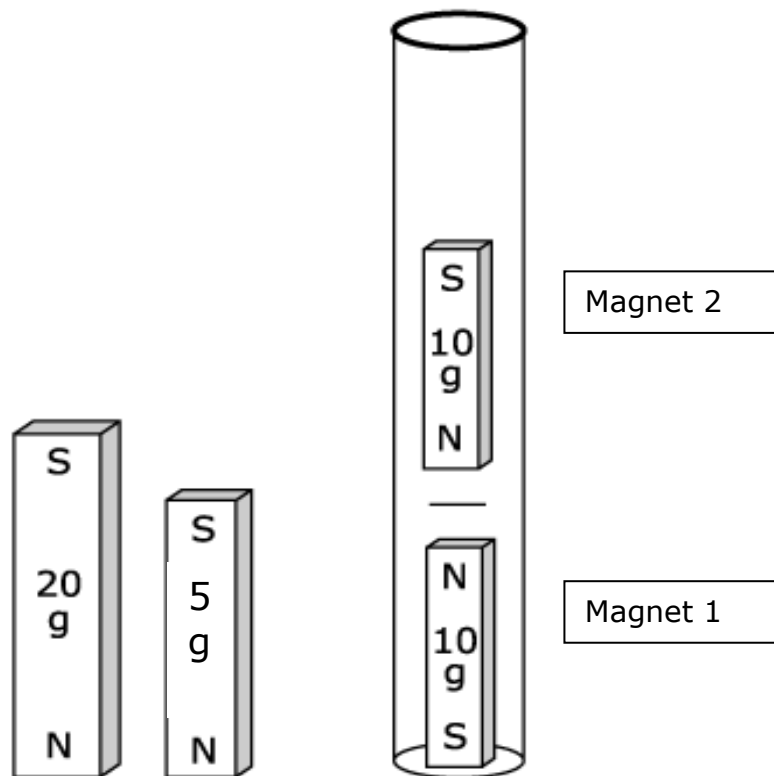
Figure 1. Bar Magnets in a Tube Experiment Setup



Magnets in a Tube – Part 2

The students plan to drop each of the other magnets into the tube, one at a time, with the north poles facing down. The students first drop the 10g magnet into the tube, and it falls, slows down, and then comes to a rest before it touches the other magnet in the tube. Figure 2 shows the results of the test with the 10g magnet.

**Figure 2. Two 10 g
Bar Magnets in a Tube**



1. Which statement **best** explains why the magnet in the tube in Figure 1 (Part 1) is not moving?
- M. A force only acts on the magnet when a second magnet is in the tube.
 - P. A force that is equal to the weight of the magnet is pushing up on the bottom of the magnet.
 - R. The only force is the weight of the magnet pushing on the bottom of the tube.
 - S. The force of the tube pushing on the magnet is greater than the force of the magnet pushing on the tube.
2. Based on Figure 2 (Part 2), which two types of forces on the magnets act at a distance?
- A. electrical
 - B. friction
 - C. gravitational
 - D. magnetic
 - E. elastic
3. Which statement **best** describes the forces in Figure 2 (Part 2)?
- M. The forces oppose the direction of an object's motion and increase as the speed of the object increases.
 - P. The forces work in the same direction as an object's motion and decrease as the speed of the object decreases.
 - R. The forces are applied to other objects through a force field and increase as the distance between objects decreases.
 - S. The forces cause air particles to apply a force on other objects and increase as the distance between objects decreases.

4. Which statement **best** explains why the dropped magnet hovers above the magnet at the bottom of the tube?
- A. The gravitational force and the magnetic force on the dropped magnet are balanced.
 - B. The gravitational force and the magnetic force on the dropped magnet are unbalanced.
 - C. Gravitational fields in the tube apply equal but opposite gravitational forces to each magnet.
 - D. The magnetic field in the tube is weaker at the bottom of the tube than at the top of the tube.
5. Which **two** statements below describe the forces acting upon the magnets in Figure 2 (Part 2) when magnet 2 is hovering above magnet 1?
- M. There is a downwards gravitational force acting on both magnet 1 and magnet 2.
 - P. There is an upwards gravitational force acting on both magnet 1 and magnet 2.
 - R. There is a downwards gravitational force acting on magnet 2 and an upward gravitational force acting on magnet 1.
 - S. There is a downwards magnetic force acting on magnet 2 and an upwards magnetic force acting on magnet 1.
 - T. There is an upwards magnetic force acting on magnet 2 and a downwards magnetic force acting on magnet 1.

6. The students set a goal to cause one of the magnets to hover at the line shown on the tube in Figure 2 (Part 2). Which magnet would **most likely** hover at the line?
- A. The 20-gram magnet would most likely hover at the line because the gravitational force acting on it would be greater than the gravitational force acting on the 10-gram magnet.
 - B. The 20-gram magnet would most likely hover at the line because the magnetic force acting on it would be greater than the magnetic force acting on the 10-gram magnet.
 - C. The 5-gram magnet would most likely hover at the line because the gravitational force acting on it would be greater than the gravitational force acting on the 10-gram magnet.
 - D. The 5-gram magnet would most likely hover at the line because the magnetic force acting on it would be greater than the magnetic force acting on the 10-gram magnet.

Metadata – Science Items

Page Number	Item Type	Key	TN Standards
1	MC	B	8.ESS2.5
2	MC	R	8.LS4.4
3	MC	B	8.LS4.1
4	MC	R	8.LS4.2
5	MC	A	8.LS4.5
6	MC	R	8.PS4.2
7	MC	D	8.ESS2.1
8	MC	S	8.ESS2.5
9	MC	C	8.PS4.1
10	MC	R	8.PS2.5

Magnets in a Tube

Page Number	Item Type	Item Number	Key	TN Standards
11	Stimulus			
12	Stimulus			
13	MC	1	P	8.PS2.4
13	MS	2	C,D	8.PS2.2
13	MC	3	R	8.PS2.2
14	MC	4	A	8.PS2.4
14	MS	5	M,T	8.PS2.2
15	MC	6	A	8.PS2.4

Metadata Definitions

Item Type	MC = Multiple Choice MS = Multiple Select Stimulus = Cluster support information
Key	Correct Answer
TN Standards	Primary standard assessed in item.